Perverse Sheaves

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1 Introduction

1.1 Overview of the Course

Given a topological space X, we can compute its singular cohomology $H^\bullet(X;k)$ over a commutative ring k.

Example 1.1.

$$H^i(\{\mathrm{pt}\};k) = \begin{cases} k, & (i=0), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Example 1.2.

$$H^i(S^1;k) = \begin{cases} k, & (i=0,1), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Definition 1.1. *Let X be a topological space. The i-th Betti number of X is*

$$b_i := \dim_{\mathbb{Q}} H^i(X;\mathbb{Q}).$$

Example 1.3. *In Example 1.2, the Betti numbers are*

$$b_i(S^1) = \begin{cases} 1, & (i=0,1), \\ 0, & (\textit{ otherwise }). \end{cases}$$

Theorem 1.1 (Poincaré Duality)**.** *Let M be a compact oriented manifold without boundary of dimension d, then for any $0 \leq i \leq d$, we have*

$$H^i(M;\mathbb{Q}) \cong H^{d-i}(M;\mathbb{Q})^\vee.$$

In terms of Betti numbers, this gives us

$$b_i(M) = b_{d-i}(M).$$

Theorem 1.2 (Hard Lefschetz)**.** *X a smooth complex projective algebraic variety of dimension $\dim_{\mathbb{C}} X = d$. Then there is an isomorphism*

$$H^i(X,\mathbb{Q}) \to H^{2d-i}(X,\mathbb{Q})$$

given by multiplication with some power of $l \in H^2(X,\mathbb{Q})$.

$$H^i(X,\mathbb{Q}) \xrightarrow{l} H^{i+2}(X,\mathbb{Q}) \xrightarrow{l} \cdots \xrightarrow{l} H^{2d-i}(X,\mathbb{Q})$$
$$\sim$$

We then get $b_i \leq b_{i+2}$ if $i < d$ and $b_i \geq b_{i+2}$ if $i \leq d$.

$$\begin{aligned} b_0 \leq b_2 \leq \cdots \leq & b_{\text{middle}} \geq \cdots \geq b_{2d}, \\ b_1 \leq b_3 \leq \cdots \leq & b_{\text{middle}'} \geq \cdots \geq b_{2d-1}. \end{aligned}$$

Definition 1.2. *For a complex projective variety X, we define*

$$H^{ij}(X) := H^i(X, \Omega_X^j).$$

Theorem 1.3.

$$H^k(X, \mathbb{C}) = \bigoplus_{i+j=k} H^{i,j}(X),\ H^{i,j} \cong \overline{H^{j,i}(X)}.$$

Furthermore,

$$b_k = \sum_{i+k} b_{i,j}, b_{i,j} = b_{j,i}.$$

Corollary 1.1. *In the situation of Theorem 1.4, we have k is odd $\Rightarrow$ b_k is even.*

Notation 1.1. *For a topologica space X, the constant sheaf of an abelian group A is denoted by $\underline{A}_X$.*

Theorem 1.4 (Deligne)**.** *For be a smooth morphism $f : X \to Y$ between smooth projective complex varieties, we have*

$$Rf_*\underline{\mathbb{Q}}_X \cong \bigoplus_{i\in\mathbb{Z}} \mathcal{H}^i(Rf_*\underline{\mathbb{Q}}_X)[-i].$$

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Remark 1.1. *Consider,*

$$S^1 \xrightarrow{f \text{ a Hopf map}} S^3 \to S^2$$

Then $Rf_*\underline{\mathbb{Q}}_{S^3}$. However, $\mathcal{H}^0(Rf_*\underline{\mathbb{Q}}_{S^3}) = \underline{\mathbb{Q}}_{S^2}$ and $\mathcal{H}^1(Rf_*\underline{\mathbb{Q}}_{S^3}) = \mathbb{Q}_{S^2}$. Therefore,

$$Rf_*\underline{\mathbb{Q}}_{S^3} \ncong \underline{\mathbb{Q}}_{S^2} \oplus \underline{\mathbb{Q}}_{S^2}[-1], H^*(S^3) \ncong H^*(S^2) \oplus H^{*-1}(S^2).$$

Hence f does not satisfy Theorem 1.4.

Question. *Over which space, does the Poincaré duality hold?*

This question led to the invention of the following notion.

Definition 1.3 (Intersection Cohomology)**.** *For a variety X over $\mathbb{C}$, we associate an mathematical object $\mathrm{IH}^\bullet(X;\mathbb{Q})$ which is called intersection cohomology.*

Theorem 1.5. *For a proper variety X over $\mathbb{C}$ of finite dimension, we have*

$$\mathrm{IH}^i(X;\mathbb{Q}) \cong \mathrm{IH}^{2d-i}(X;\mathbb{Q})^\vee.$$

Theorem 1.5 proves some versions of Theorem 1.2.

Notation 1.2. *Let X be a topological space. The derived category of constructible sheaves on X is denoted by $D_c^b(X)$.*

Theorem 1.6. *For a proper variety X over $\mathbb{C}$ of finite dimension, there is a complex of sheaves $\mathrm{IC}_X \in D_c^b(X)$ such that*

$$\mathrm{IH}^*(X, \mathbb{Q}) \cong H^*(X, \mathrm{IC}_X).$$

Definition 1.4. *Such IC_X in Theorem 1.6 is called a perverse sheaf.*

Remark 1.2. *Perverse sheaves were introduced by Belinson-Bernstein-Deligne in 1983. Together with Gabber, they extended Theorem 1.4 to the case when $f : X \to Y$ is proper.*

1.2 Preliminaries

Notation 1.3. *Let X be a topological space and k be a commutative ring. The category of presheaves of k-modules over X is denoted by $\mathrm{PreShv}(X;k)$, similarly the sheaves of k-modules over X is denoted by $\mathrm{Shv}(X;k)$.*

Example 1.4. *For all $M \in \mathrm{Mod}(k)$, there is a constant sheaf $\underline{M}_X \in \mathrm{Shv}(X;k)$.*

Definition 1.5. *A local system on a topological space X is a sheaf on X which is locally a constant sheaf of an abelian group.*

Example 1.5. *For $r > 0$, consider morphisms of groups,*

$$p : \pi_1(X, x_0) \to \mathrm{Aut}(k^{\oplus r}).$$

Define L_p as follows. Let $\pi : \tilde{X} \to X$ be the universal covering space. For $U \subseteq X$,

$$L_p := \{\pi^{-1}(U) \to k \mid \forall g \in \pi_1(X, x_0), f(x \cdot g) = p(g)^{-1} f(x)\}.$$

Then L_p is a local system on X.

Theorem 1.7. *$\mathrm{Shv}(X;k)$ is an abelian category.*

Proposition 1.1. *The canonical inclusion $\mathrm{Shv}(X;k) \to \mathrm{PreShv}(X;k)$ admits a left adjoint and such functor is called the sheafification,*

$$\mathrm{PreShv}(X;k) \ni \mathcal{F} \mapsto \mathcal{F}^\dagger \in \mathrm{Shv}(X;k).$$

Explicitly, for $\mathcal{F} \in \mathrm{PreShv}(X;k)$ and $\mathcal{G} \in \mathrm{Shv}(X;k)$, we have

$$\mathrm{Mor}_{\mathrm{PreShv}}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{G}),$$

which is natural. In particular, setting $\mathcal{G} = \mathcal{F}^\dagger$, there is a natural map $\mathcal{F} \to \mathcal{F}^\dagger$ which is obtained from $\mathrm{id} \in \mathrm{Mor}_{\mathrm{Shv}}(\mathcal{F}^\dagger, \mathcal{F}^\dagger)$.

Proposition 1.2. *The natural morphism $\mathcal{F} \to \mathcal{F}^\dagger$ induced isomorphisms on stalks.*

Definition 1.6. *Given a complex of sheaves,*

$$\mathcal{F}^\bullet := \cdots \to \mathcal{F}^{i-1} \overset{d^{i-1}}{\to} \mathcal{F}^i \overset{d^i}{\to} \mathcal{F}^{i+1}.$$

we define its cohomology of sheaves to be

$$\mathcal{H}(\mathcal{F}^\bullet) := \ker d^i \oplus \mathrm{im}\, d^{i-1}.$$

A morphism of complexes is called a quasi-isomorphism if it induces isomorphisms in cohomology.

Definition 1.7. *The derived category on $\mathrm{Shv}(X;k)$ is a category satisfying the following.*

i. The objects are complex of sheaves,

$$\mathcal{F}^\bullet := \cdots \to \mathcal{F}^{i-1} \overset{d^{i-1}}{\to} \mathcal{F}^i \overset{d^i}{\to} \mathcal{F}^{i+1}.$$

ii. The morphisms are morphisms of complexes.

iii. Quasi-isomorphisms are invertible.

Remark 1.3. *The derived category of sheaves on X over k is a triangulated category.*

However the derived category turns out to be to large to handle. Therefore, we will use a smaller category called the bounded constructible derived category of sheaves denoted by $D^b_c(X;k)$.

Remark 1.4. *The category of bounded constrictible derived category of sheaves $D^b_c(X)$ is a subcategory of the derived category of sheaves which is more manageable.*

Definition 1.8. *There exists a stratification in locally closed subspaces $X = \bigsqcup_\alpha S_\alpha$. That is for $\mathcal{F}^\bullet \in D_c^b(X)$, we have $\mathcal{H}^i(\mathcal{F}^\bullet)|_{S_\alpha}$ is a finite rank local system and $\mathcal{H}^i(\mathcal{F}^\bullet) = 0$ for $|i| \gg 0$.*

Remark 1.5. *Let $f : X \to Y$ be a continuous map. The following so-called six-functors are important when it comes to produce example of sheaves,*

$$Rf_* \dashv f^*, f_! \dashv f^!, (\cdot \otimes \cdot) \dashv \mathrm{Hom}(\cdot, \cdot).$$

Remark 1.6. *If f is proper then $Rf_* = f_!$, if f is open immersion then $f^* = f^!$.*

Theorem 1.8.

$$D_c^b(X; k) \cong D^b(\mathrm{Shv}_c(X; k)).$$

Example 1.6. *For a scheme X and $\alpha : X \to \mathrm{pt}$, we have,*

$$\alpha_* \underline{k}_{\mathrm{pt}} \cong \underline{k}_X, \alpha^! \underline{k}_{\mathrm{pt}} \cong \omega_X.$$

With the aid of these functors, we define(reconstruct) following complexes.

Definition 1.9.

$$\begin{aligned} H^\bullet(X; k) &:= \alpha_* \alpha^* \underline{k}_{\mathrm{pt}}, \\ H^{\mathrm{BM}}(X; k) &:= \alpha_* \alpha^! \underline{k}_{\mathrm{pt}}, \\ H_c^*(X; k) &:= \alpha_! \alpha^* \underline{k}_{\mathrm{pt}}, \\ H_*(X; k) &:= \alpha_! \alpha^! \underline{k}_{\mathrm{pt}}. \end{aligned}$$

Example 1.7. *Let $Z \subseteq X$ be closed and $U = X \backslash Z$. Consider $Z \overset{i \text{ closed}}{\hookrightarrow} X \overset{j \text{ open}}{\hookleftarrow} U$. This gives natural transformations,*

$$(\cdot) \to i_* i^*(\cdot), (\cdot) \to j_* j^*(\cdot), (\cdot) \to i_! i^!(\cdot), (\cdot) \to j_! j^!(\cdot).$$

Furthermore, this gives an exact triangle,

$$j_! j^!(\cdot) \to (\cdot) \to i_* i^*(\cdot) \to [+1].$$

Using Remark 1.6 and plugging in $\underline{k}_X = \alpha_* \underline{k}_{\mathrm{pt}}$, we obtain

$$\alpha_* j_! \underline{k}_U \to \alpha_* \underline{k}_X \to \alpha_* i_* \underline{k}_Z.$$

Example 1.8. *In Example 1.7, we also have another triangle*

$$i_! i^!(\cdot) \to (\cdot) \to j_* j^*(\cdot) \to [+1].$$

Plug $\omega_X := \alpha^! \underline{k}_\pi$ to it and use Remark 1.6, we obtain

$$i_* \omega_x \to \omega_x \to j^* \omega \to [+1].$$

Applying α_*, we get

$$\alpha_* i_* \omega_x \to \alpha_* \omega_x \to \alpha_* j^* \omega \to [+1].$$

This gives a rise to the following long exact sequence,

$$\cdots \to H_i^{\mathrm{BM}}(Z; k) \to H_i^{\mathrm{BM}}(X; k) \to H_i^{\mathrm{BM}}(U; k) \to \cdots$$

Example 1.9. *Let $X = U \cup V$ be a covering and $W = U \cap V$. Set j_U, j_V, j_W be the corresponding canonical inclusions. Then*

$$(\cdot) \to j_{U,*} j_U^*(\cdot) \oplus j_{V,*} j_V^*(\cdot) \to j_{W,*} j_W^*(\cdot) \to [+1].$$

Plug $\underline{k}_X$ and apply α_*, we obtain Mayer-Vietoris,

$$H^\bullet(X; k) \to H^\bullet(U; k) \oplus H^\bullet(V; k) \to H^\bullet(W; k) \to H^{\bullet+1}(X; k) \to \cdots$$

Suppose $X = U \cup V$ and $W = U \cap V$, and there are natural transformations,

$$(\cdot) \to (j_V)_* (j_U)^* (\cdot).$$

1.3 Informal Definition of Perverse Sheaves

Definition 1.10.

$$\mathbb{D}(\cdot) = \mathrm{Hom}(\cdot, \omega_X) : \mathrm{Shv}(X;k) \to \mathrm{Shv}(X;k).$$

Notation 1.4. *We will consider* $\mathrm{Shv}(X;k)$ *in its derived category. The full subcategory of constructible sheaves is denoted by* $\mathrm{Shv}_c(X;k)$*. The constructible bounded sheaves is then denoted by* $D^b_c(X;k) = D^b(\mathrm{Shv}_c(X;k))$*.*

$$\begin{array}{ccc} \mathrm{Shv}(X;k) & \hookrightarrow & D^b(\mathrm{Shv}(X;k)) \\ \uparrow & & \uparrow \\ \mathrm{Shv}_c(X;k) & \hookrightarrow & D^b_c(\mathrm{Shv}(X;k)) \end{array}$$

Definition 1.11. *Let* X *be a variety of finite type over a field* k*.* $\mathcal{F} \in D^b_c(X;k)$ *is a perverse sheaf if for all* $n \in \mathbb{Z}$*,* $\dim_k \mathrm{supp}\, \mathcal{H}(\mathcal{F}) \leq i$ *and* $\dim \mathrm{supp}\, \mathcal{H}(\mathbb{D}(\mathcal{F})) \leq i$*.*

The full subcategory of perverse sheaves is denoted by $\mathrm{Perv}(X;k) \subseteq D^b_c(\mathrm{Shv}(X;k))$*. This turns out to be an abelian category.*

Example 1.10. *Let* X *be a smooth variety over* $\mathbb{C}$ *of dimension* d*,*

$$\omega_X = k_X[2d].$$

Then $\underline{k}_X[d]$ *is a perverse sheaf and* $\mathbb{D}(\underline{k}_k[d]) = \underline{k}_X[d]$*. As,*

$$\mathcal{H}(k_x[d]) = \begin{cases} \underline{k}_X & (i = d), \\ 0, & (\ \textit{otherwise}\ ). \end{cases}$$

Hence

$$\dim \mathrm{supp}(\mathcal{H}(\underline{k}_X[d])) = \begin{cases} d, & (i = d), \\ -\infty, & (\ \textit{otherwise}\ ). \end{cases}$$

Similarly, if L *is a local system on* X*, then* $L[d] \in \mathrm{Perv}(X;k)$*.*

Theorem 1.9 (Bellinson-Bernstein-Deligne)**.** $\mathrm{Perv}(X;k)$ *is an Abelian category which is both Noetherian and Artinian. Furthermore,* $D^b(\mathrm{Perv}(X;k)) = D^b(X;k)$*.*

Remark 1.7. *Every perverse shaef as a composition series where all objects are simple.*

Hence it is an interesting question to ask how these simple objects are constructed.

Remark 1.8. *Consider a smooth variety of finite type* X*,* Z *a closed subvariety and* V *an open smooth subset contained in* Z*.*

Let L *be a local system on* V*, then one can construct a perverse sheaf* $\mathrm{IC}_Z(L) \in \mathrm{Perv}(X;k)$ *which is supported on* Z *and* $\mathrm{IC}_Z(L)|_V = L[\dim V]$*.*

Example 1.11. *Let* X *be a variety of finite type and* $U = X^{\mathrm{sm}}$ *be an smooth open subvariety.* $L = \underline{k}_U$ *is a local system and we call* $\mathrm{IC}_X := \mathrm{IC}_X(\underline{k}_U)$ *the intersection complex. This turns out to be isomorphic to* $\underline{\mathbb{Q}}_X[\dim X]$ *whose cohomology is* $IH^\bullet(X;k)$*. This satisfies both Poincaré duality and hard Lefschetz.*

Recall Theorem 1.4 and the formula

$$\underbrace{Rf_*\underline{\mathbb{Q}}_X}_{\in D^b_c(Y;k)} \cong \bigoplus_{i\in\mathbb{Z}} \mathcal{H}^i(\underbrace{Rf_*\underline{\mathbb{Q}}_X}_{\in \mathrm{Shv}_c(Y;k)})[-i].$$

where $f : X \to Y$ is a smooth projective map between smooth projective varieties. We have a stronger statement to it.

Theorem 1.10. *For a proper map $f : X \to Y$ between varieties of finite over k, we have*

$$\underbrace{f_*\mathrm{IC}_X}_{D^b(\mathrm{Perv}(Y;k))=D^b(Y;k)\ni} \cong \bigoplus_{i\in\mathbb{Z}} {}^p\mathcal{H}^i(\underbrace{f_*\mathrm{IC}_X}_{\in\mathrm{Perv}(Y;k)})[-1],$$

where for all $i \in \mathbb{Z}$, we set

$${}^p\mathcal{H}^i(f_*\mathrm{IC}_X) = \bigoplus_j \mathrm{IC}_{Z_j}(L_j),$$

where each summand on the right hand side is simple.

Remark 1.9. *I the derived cateogory, $f_*\mathrm{IC}_X$ admits is quasi-isomorphic to a complex with each term a perverse sheaf. Hence it makese sense to talk about the above cohomology.*

Remark 1.10. *Let $f : X \to Y \xrightarrow{a}$ pt be a resolution of singularities. From Theorem 1.4 and Theorem 1.10,*

$$\underbrace{f_*\mathrm{IC}_Y}_{\cong f_*\underline{\mathbb{Q}}[\dim Y]} = \bigoplus_{\substack{\text{closed subvarieties } Z_i \\ \text{local systems } L_i, \\ \text{on open subset of } Z_i}} \mathrm{IC}_Z(L_i)[\textit{shifts}]$$

Furthermore IC_X appears as a summand on the right hand side.

Using that f is a resolution of singularities hence induces an isomorphism $f^{\mathrm{sm}} : X^{\mathrm{sm}} \xrightarrow{\sim} Y^{\mathrm{sm}}$. Therefore, we obtain

$$f_*\mathrm{IC}_Y|_{X^{\mathrm{sm}}} = (f^{\mathrm{sm}})_*\mathrm{IC}_{Y^{\mathrm{sm}}} = \underline{\mathbb{Q}}_{X^{\mathrm{sm}}}[\dim X].$$

Hence there will be a summand of $f_*\mathrm{IC}_Y$ which restricts to $\underline{\mathbb{Q}}_{X^{\mathrm{sm}}}$ which will turn out to be IC_X. Furthermore, by applying a_*, $\mathrm{IH}^\bullet(X)$ will appear as a summand of $H^\bullet(Y;k)$.

References

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[2] Mark de Cataldo, Luca Migliorini, The Decomposition Theorem and the topology of algebraic maps.